

Inputs

Building stock

footprints, archetypes,
dwelling counts

Costs and performance

capital, O&M, efficiencies,
lifetimes, discount rate

Power and geology

2050 capacity, renewables,
salt caverns



Heat demand

Residential useful heat reconstructed bottom-up for 1,369 NUTS3 regions
across 29 markets, validated against national statistics



Scenarios

200 Monte Carlo draws over 2025 to 2050, under Current Policies,
Stated Policies, Net Zero and H2 Push



Three tests, applied in order

1. Levelised cost

Which carrier gives a MWh
of useful heat for least,
each charged its last mile
and its seasonal store?



2. Merit order

Which is cheapest to run
at the winter peak, across
building heat, district
heat and the power peaker?



3. Capital recovery

Does scarcity rent cover
the annualised cost of
building the plant, and if
not, what payment would?



Outputs

Cost gaps and technology shares by country, emissions pathways, and the
capacity payment a hydrogen peaker would need to be built